Predicting Patient Drop-off

Abstract

Flagler Health provides Remote Therapeutic Monitoring (RTM) for patients with musculoskeletal pain. This study addresses the research question: What factors can predict patient disengagement from the RTM program? The objective is to identify patterns in clinical and behavioral data to inform a dynamic risk score for patient drop-off. Data from 391 patients over the last 4 months of 2025. Covariates used are phone check-in metrics (pain, mobility, mood, sleep), message transcripts, and insurance codes. The outcome variable is stripped from status change logs and is used to evaluate the models. We performed data cleaning and integration using SQL (DuckDB) to translate the original json files into csv files. We used different methods to evaluate the dataset, including individual-level longitudinal EDA with annotated pain trajectories; baseline pain score distribution comparison between enrolled and disenrolled patients; Cox proportional hazards and logistic regression models predicting dropout using baseline pain, mobility, mood, sleep, and procedure and diagnosis groups.

Baseline pain distributions were broadly similar between groups, with no statistically significant predictors in the Cox model. However, the procedure group was significant: the Other/Surgery group had a lower dropout hazard compared to the Evaluation & Management group. Logistic regression confirmed this direction of effect.

The procedure grouping, specifically the distinction between evaluation/management visits and others (surgery-related care, radiology), showed consistent associations with engagement. Patients with surgery-related codes may have stronger long-term care relationships, reducing dropout. While diagnosis grouping yielded statistically weaker results than procedure grouping when examined separately, we are now combining both to develop a more robust predictive model, including creating four combination groups from diagnosis and procedure categories to observe trends via Kaplan-Meier curves. The inclusion of sentiment further suggests that behavioral signals may complement administrative and clinical variables by capturing patient-level variation in engagement. Next steps include deeper text analysis (e.g., sentiment over time, last-message sentiment, and response rates) and logistic regression with repeated-measure metrics. Overall, these findings suggest that administrative and behavioral data, not just clinical metrics, may improve prediction of patient disengagement.

Introduction

Flagler Health is a healthcare technology startup that provides Remote Therapeutic Monitoring (RTM) services for patients with musculoskeletal pain, such as back pain or knee pain. RTM is a program introduced by the Centers for Medicare and Medicaid Services (CMS) that encourages providers to track non-physiological, behavioral health data between hospital visits by reimbursing them for monitoring that data. It aims to improve treatment adherence and health outcomes by helping providers monitor progress in real time and adjust care as needed.

Flagler's RTM service collects patient data to track how a patient's condition changes over time and to help providers meet insurance billing requirements for treatments and surgeries. A key goal of the company is to enroll patients into the program and keep them actively engaged. This project analyzes patient data and interactions with medical assistants to identify patterns in patient engagement and understand when and why patients drop out of the program. For the scope of our research question, we are focused on the Remote Therapeutic Monitoring (RTM) project under the company, and only taking patients enrolled under this program into consideration when doing our analysis.

Our research question is: What factors are associated with patient disengagement? While we initially explored multiple modeling approaches, our final model is a logistic regression that produces probability outputs, which can be translated into binary classifications using a chosen threshold.

Methods

The primary types of data provided for this project fall into four categories: metrics, status change, messages, and insurance.

The data collection process involved medical assistants conducting phone check-ins with 391 patients between August and December 31, 2025. During each call, patients or assistants completed structured electronic forms that recorded quantitative health indicators, thereby generating chat data linked to each interaction. The key metric variables extracted from these forms include sleep, mobility, and mood. Also a continuous measure of pain level (0-10). Linking variables, date, chat_id, and patient_id, allow these metric data to be connected with other datasets for longitudinal or cross-sectional analysis.

Status change data track changes in a patient's program status. These records are stored as JSON files, with one file per patient containing a timeline of status updates. For this analysis, we filtered the data to include only two relevant status changes: "RTM to Enrolled" and "RTM to Declined." From these records, we created additional variables such as enrollment status and number of days enrolled.

Messages data contain the longitudinal communication history between patients and medical assistants. These messages include both casual conversation and health-related discussions. In addition to message text, some conversations include reported health metrics such as the Numeric Pain Rating Scale (NPRS, 0–10), mobility level (Not Active/Active/Very Active), mood (Sad/Flat/Energetic), and sleep quality (Poor/Moderate/Restful).

The insurance data capture patients' medical billing history, including service dates, procedure codes, diagnosis codes, and diagnosis code position numbers. The main variables of interest are the procedure and diagnosis codes, which help to provide an overview of each patient's medical history. These codes follow standardized insurance coding systems: CPT/HCPCS for procedures and ICD-10 for diagnoses. For example, the CPT procedural code 99204 indicates a high-level

45-60 minute office visit for a new patient, while the ICD-10 diagnosis code Z79.891 indicates long-term use of opioid analgesics.

Step #1: Data Cleaning

To prepare the data for analysis, we first conducted a data cleaning and integration process to combine information from multiple sources into structured tables. The raw data were stored primarily as JSON files, with some datasets provided as a single unified file and others stored as separate files for each patient. To efficiently process and query these files, we used DuckDB through its Python interface (DuckPy) to run SQL queries directly on the JSON data.

Using SQL queries, we extracted relevant variables and standardized the data structure across files. We then combined datasets using common identifiers such as patient_id, date, and chat_id to create integrated tables suitable for analysis. During this process, we also created additional variables needed for the project, such as indicators for enrollment status, enrollment and disenrollment dates, and derived measures like days enrolled in the program.

After extracting and structuring the relevant fields, the cleaned datasets were exported to CSV format for easier downstream analysis. This workflow allowed us to systematically transform the raw JSON data into organized tables that link patient metrics, communication records, status changes, and insurance information.

Step #2: Individual Exploratory Data Analysis

To better understand how patient engagement evolves over time, we conducted an individual-level exploratory data analysis focusing on the longitudinal health metrics of specific patients. The goal of this analysis was to visually examine how patient-reported health indicators and communication events relate to continued enrollment or program drop-off.

First, we selected individual patients from the dataset and extracted their recorded health metrics over time. In particular, we focused on the Numeric Pain Rating Scale (NPRS), which ranges from 0 to 10. For each selected patient, we filtered the metrics dataset by patient_id and removed observations where the pain score was missing. The removal wasn't significant, as only 7 out of 391 observations were removed. The date variable was converted into a standard date format to allow for time-series visualization.

To compare engagement patterns, we produced separate visualizations for two example patients: one who remained enrolled in the program and another who eventually dropped out. We created line plots showing each patient's pain score across the observation period. We then highlighted specific dates corresponding to important events in the patient's timeline, such as enrollment in the program, notable reported symptoms, or the date the patient chose to leave the program. Messages from the text dataset that corresponded to these dates were manually extracted and included as annotations on the plots. This allowed us to link quantitative health metrics with qualitative patient feedback and contextual events. By examining the trajectories of their pain scores alongside the messages, we were able to explore potential differences in health progress and engagement behavior between retained and disengaged patients.

Step #3: Baseline Pain Score Analysis

To examine whether initial pain levels differ between patients who remained enrolled and those who eventually disengaged from the program, we conducted an exploratory analysis of baseline (first day of enrollment) Numeric Pain Rating Scale scores across enrollment groups.

We first identified each patient's baseline pain score, defined as the earliest recorded NPRS value associated with their patient identifier. Patients were then categorized into two groups based on program status: those who remained enrolled and those who later disenrolled from the program. For each group, we calculated the frequency distribution of baseline pain scores and converted these counts into percentages to allow for direct comparison between groups of different sizes.

To visualize these distributions, we constructed a bar chart displaying the percentage of patients in each NPRS category (0–10) for both enrollment groups. Bars for enrolled and disenrolled patients were plotted side-by-side to facilitate comparison of pain score frequencies across the two groups. This visualization allowed us to assess whether patients who eventually left the program exhibited different baseline pain patterns compared to those who remained engaged.

By comparing the distributions, we were able to explore whether higher or lower initial pain levels were associated with program retention or disengagement, providing preliminary insight into how baseline health status might relate to patient engagement outcomes.

Step #4: Insurance Code Categorization

To incorporate patients' medical billing history into the analysis, we categorized the insurance data into broader diagnostic and procedural groupings. The insurance dataset includes service dates, diagnosis codes, diagnosis code position numbers, and procedure codes recorded from medical claims. Since a single medical claim can include multiple diagnosis codes for the same visit or service date, we used the diagnosis code position number to focus on the primary condition for each encounter. This variable ranges from 1 to 4, with 1 indicating the main diagnosis, so we restricted the analysis to records where the diagnosis code position number was 1.

Procedure codes come from three main coding systems: CPT Category I, CPT Category II, and HCPCS. CPT I codes represent standard medical procedures and service and were the most common and informative in the dataset. These five-digit codes were grouped into 6 main clinical sections: Evaluation and Management, Anesthesia, Surgery, Radiology, Pathology and Laboratory, and Medicine (see Appendix A). CPT II codes are primarily used for qualitative measurement (i.e. 3074F: Systolic blood pressure less than 130 mmHg for a hypertensive patient) and HCPCS codes mainly identify supplies and equipment such as ambulance services and prosthetics. Since CPT II and HCPCS codes were relatively sparse and less informative for this analysis, we filtered them out and retained only CPT I codes.

Diagnosis codes follow the ICD-10 system and are grouped into 22 broader diagnostic categories based on their main code groupings (see Appendix A). Together, these diagnosis and procedure groupings allowed us to summarize patients' primary diagnoses and patterns of care in a simple and interpretable form.

Step #5: Survival Analysis

To examine whether baseline patient characteristics were associated with program disengagement over time, we conducted a survival analysis using each patient's baseline values for pain, sleep, mood, mobility, procedure group, and diagnosis group. Metric observations were missing in some rows, so the baseline was extracted separately for each variable as the first value recorded after the enrollment date. For procedure and diagnosis, the baseline value was defined as the observation closest to enrollment, which could occur either before or after the enrollment date, and ties were broken by selecting the value recorded before enrollment. These baseline variables were then merged into a single patient-level dataset using patient id.

For the survival analysis, we define the event of interest as dropout. Patients who did not drop out by December 31, 2025 are considered censored observations. We first used Kaplan-Meier curves to estimate the distribution of the time to drop out. Separate Kaplan-Meier analyses were conducted for baseline pain, sleep, mood, mobility, procedure, and diagnosis group, with baseline pain grouped into quartiles for comparison. The groups within each variable were compared using log-rank tests.

We then fit Cox proportional hazards models to assess the association between baseline variables and time to dropout. For pain, we included both a centered linear term and a quadratic term to test for possible nonlinearity. We also explored interaction terms among baseline predictors. To further examine procedure, we simplified the original procedure categories because the counts were highly imbalanced, with most observations concentrated in Evaluation and Management and Surgery. We first collapsed the six CPT I sections into two groups, Evaluation and Management versus Other, where other represents more intensive or specialized procedures. We also tested a three-group version of the procedure variable consisting of Evaluation and Management, Surgery, and Other. Similarly, for the diagnosis group, we addressed imbalance by collapsing categories into two groups: Musculoskeletal and Other.

Finally, we fit a logistic regression model predicting whether a patient drops out as a robustness check alongside the Cox model, which focuses on time to dropout. This allowed us to compare whether the same baseline variables were associated with dropout status overall as well as dropout timing.

Step #6: Sentiment Analysis of Patient-MA Messages

To incorporate the qualitative patient-MA message data into the predictive analysis, we conducted sentiment analysis on the message transcripts. Since patient engagement may be influenced not only by clinical metrics but also by the emotional tone of communication, we wanted to create a structured sentiment variable that could be included in the logistic regression model.

We used the Flair NLP (Natural Language Processing) library for sentiment classification. This choice was guided by Aggarwal et al. (2025), who compared VADER, BERT, and Flair for sentiment analysis of healthcare-related patient reviews of pain management physicians. Their study found that Flair had the strongest correlation with patient ratings, outperforming both BERT and VADER. Since their research also focused on patient language in a healthcare setting, we considered Flair a suitable tool for capturing sentiment in our patient-MA message data.

Flair outputs two main pieces of information for each message: a sentiment label, either positive or negative, and a confidence score. However, because we wanted a continuous numerical predictor that captured the magnitude of sentiment rather than only a binary label, we transformed Flair's output into a signed sentiment score for the regression model. Specifically, the confidence score was multiplied by 1 if the message was labeled positive and by -1 if the message was labeled negative. As a result, the final sentiment variable ranged from strongly negative values to strongly positive values, allowing us to preserve both the direction and strength of the sentiment.

After creating the sentiment score, we incorporated it into a logistic regression model predicting patient dropout. To evaluate whether sentiment improved the model, we ran three robustness checks. First, we compared the logistic regression model with sentiment to a reduced model without sentiment using an ANOVA model comparison. This tests whether adding sentiment as a predictor adds meaningful predictive value to the model. Second, we compared the Akaike Information Criterion (AIC) values of the two models to assess whether adding sentiment improved model fit after accounting for model complexity. Third, we performed the Hosmer-Lemeshow goodness-of-fit test to evaluate how well the logistic regression model's predicted probabilities matched the observed dropout outcomes, or its calibration. Together, these checks allowed us to assess whether sentiment added meaningful predictive value beyond the structured clinical and administrative variables already included in the model.

Results

Individual Exploratory Data Analysis

Figure 1

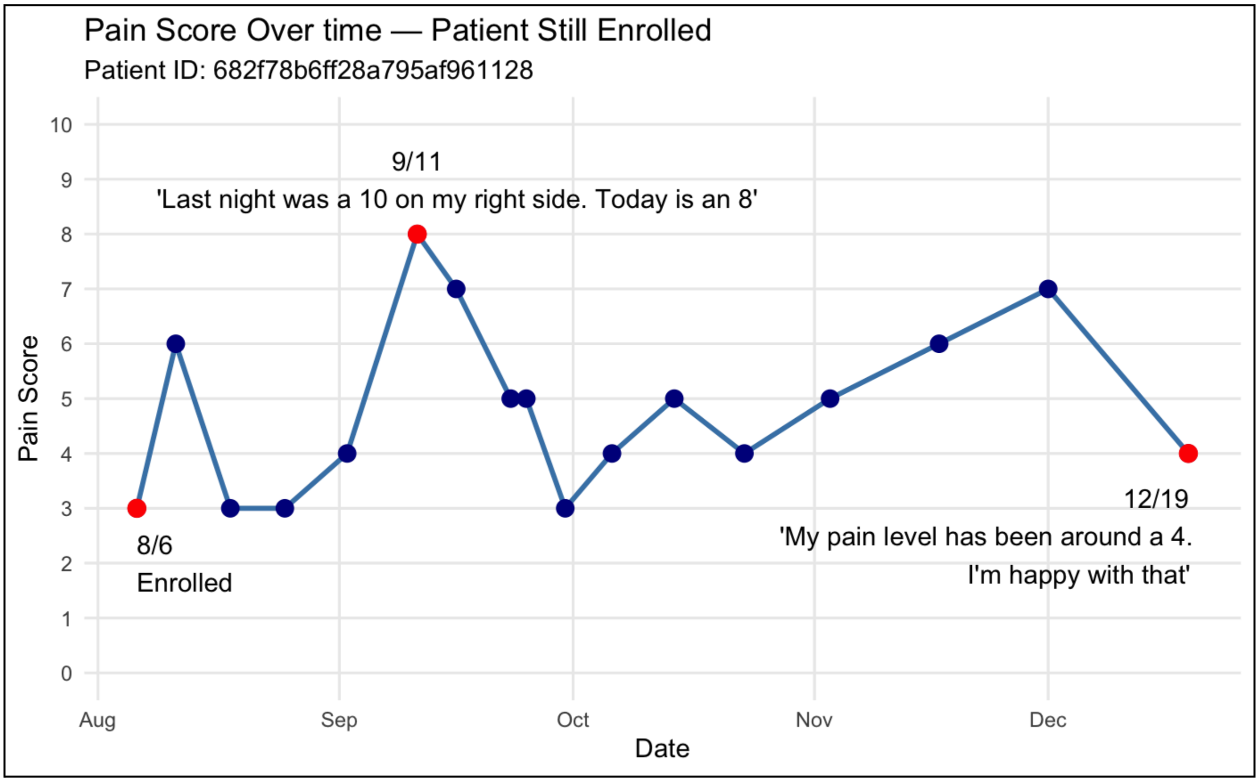


Figure 1 shows that for Patient 682f78b6ff28a795af961128, who remained enrolled in the program, pain scores exhibited significant volatility but trended toward a moderate level over four months. Following an enrollment score of 3 on August 6, the patient experienced a peak pain level of 8 on September 11, noting, "Last night was a 10... Today is an 8". Despite this peak, the patient remained engaged with frequent check-ins, eventually reaching a pain score of 4 by December 19, the last check-in date recorded. The patient's qualitative feedback at the end of the period ("I'm happy with that") suggests that continued enrollment may be linked to a patient's satisfaction with a manageable pain level, even if the pain is not entirely eliminated.

Figure 2

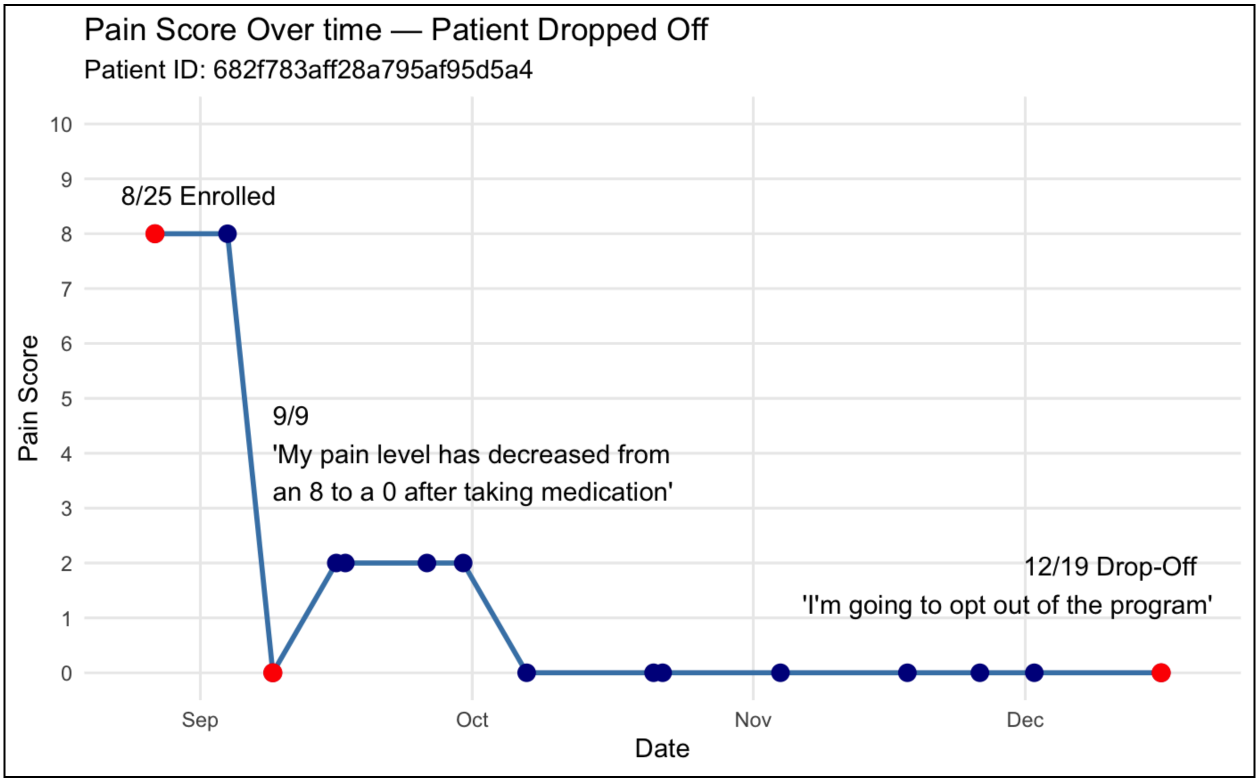


Figure 2 shows that for Patient 682f783aff28a795af95d5a4, who dropped off, pain scores plummeted rapidly and remained at zero for an extended period. This patient entered the program with a high pain score of 8 on August 25 but reported a score of 0 by September 9, stating, "My pain level has decreased from an 8 to a 0 after taking medication". Following this total reduction in pain, the patient maintained a score of 0 through October, November, and December, showing very little variation. The patient eventually disenrolled on December 19, stating, "I'm going to opt out of the program". This suggests that a lack of reported pain may actually be a risk factor for disengagement, as the patient may no longer perceive a need for RTM services.

Taken together, these two examples highlight different patterns of engagement. The patient who remained enrolled experienced fluctuating but persistent pain and continued reporting health metrics over time. In contrast, the patient who dropped out experienced a rapid reduction in pain and maintained consistently low scores prior to leaving the program. While these observations are based on individual cases and cannot establish causal relationships, they illustrate how pain trajectories and patient-reported experiences may relate to continued engagement or disengagement in the RTM program.

Baseline Pain Score Analysis

Figure 3

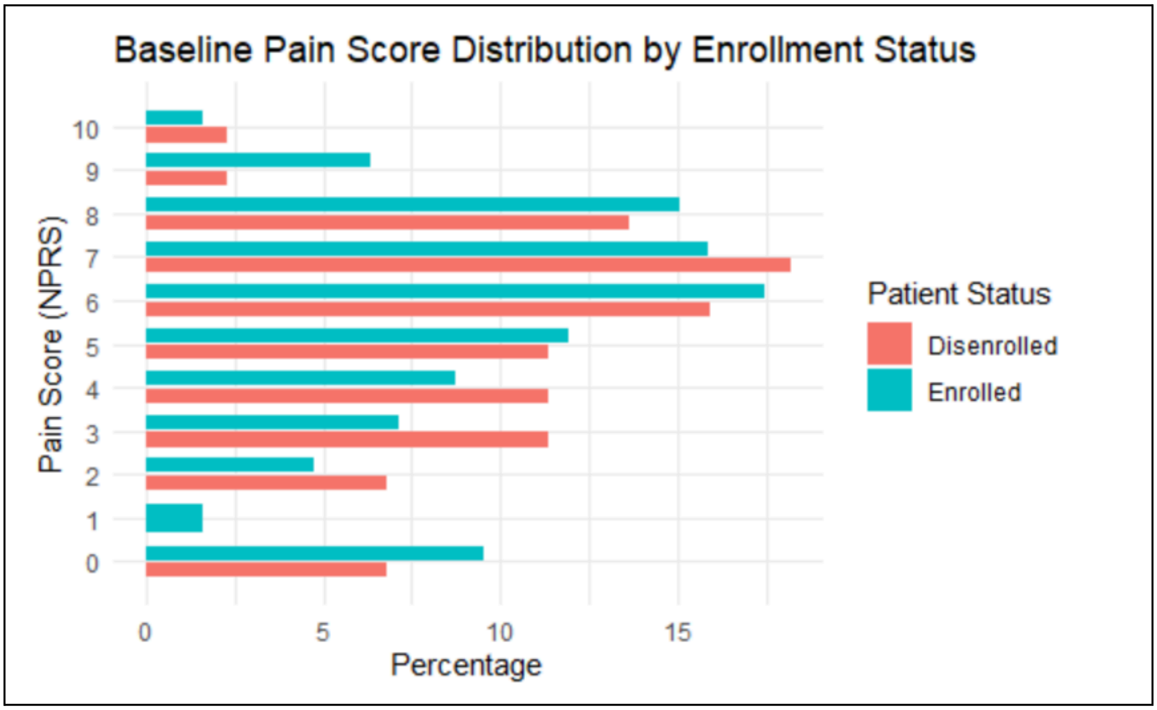


Figure 3 displays the baseline pain score distributions for enrolled and disenrolled patients. The two distributions appear broadly similar, with both groups showing the highest proportions around NPRS scores of 6-8. However, there are also some notable differences around those with NPRS scores of 2-4, with patients who eventually disenrolled show slightly higher percentages in pain score. The opposite seems to be true for those patients with NPRS scores of 5-9, with patients who eventually disenrolled show slightly lower percentages in pain score. Although there are varying patient experiences at the start of the program, we don't have strong evidence that these varieties influence a patient's enrollment status, and further analysis has to be conducted.

Survival Analysis of Baseline Variables

Figure 4

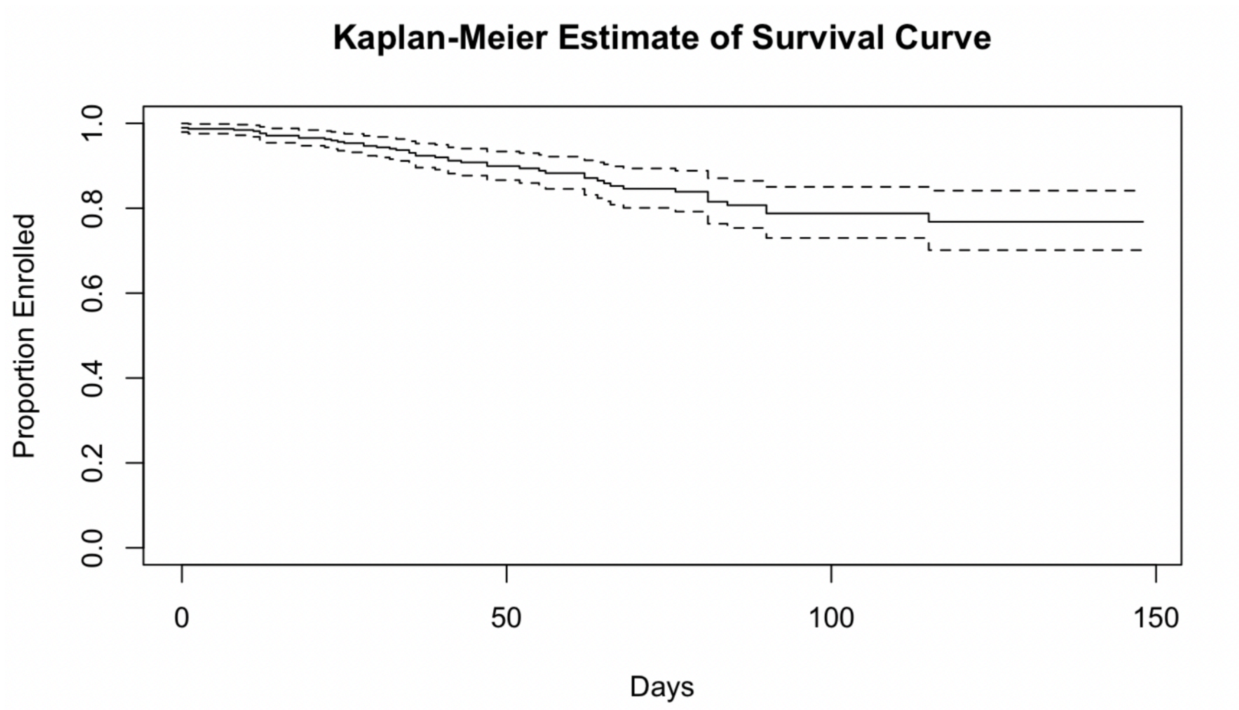


Figure 4, the overall Kaplan-Meier survival curve shows that the probability of remaining enrolled in the RTM program decreases over time, indicating that patient drop-off accumulates as the observation period progresses. However, at the end of the study period, approximately 80% of the patients were still enrolled. This implies that although disengagement is present, a majority of patients remained enrolled through the end of follow-up.

Figure 5

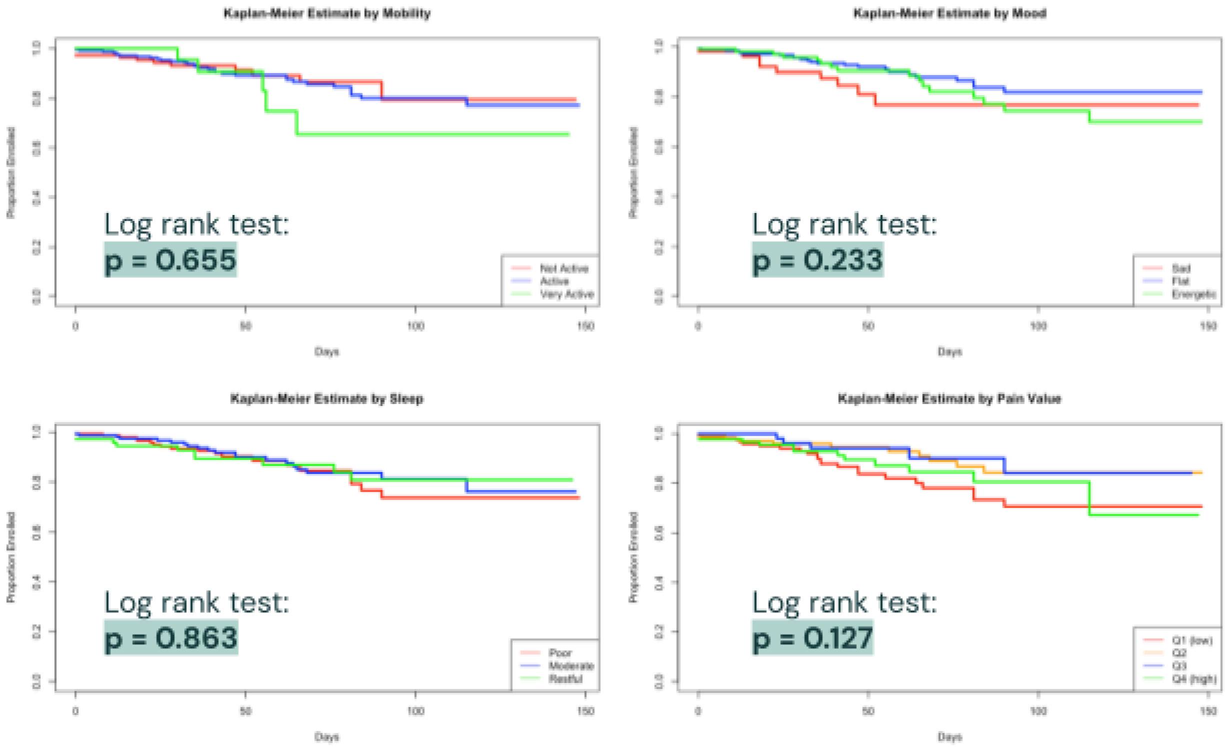
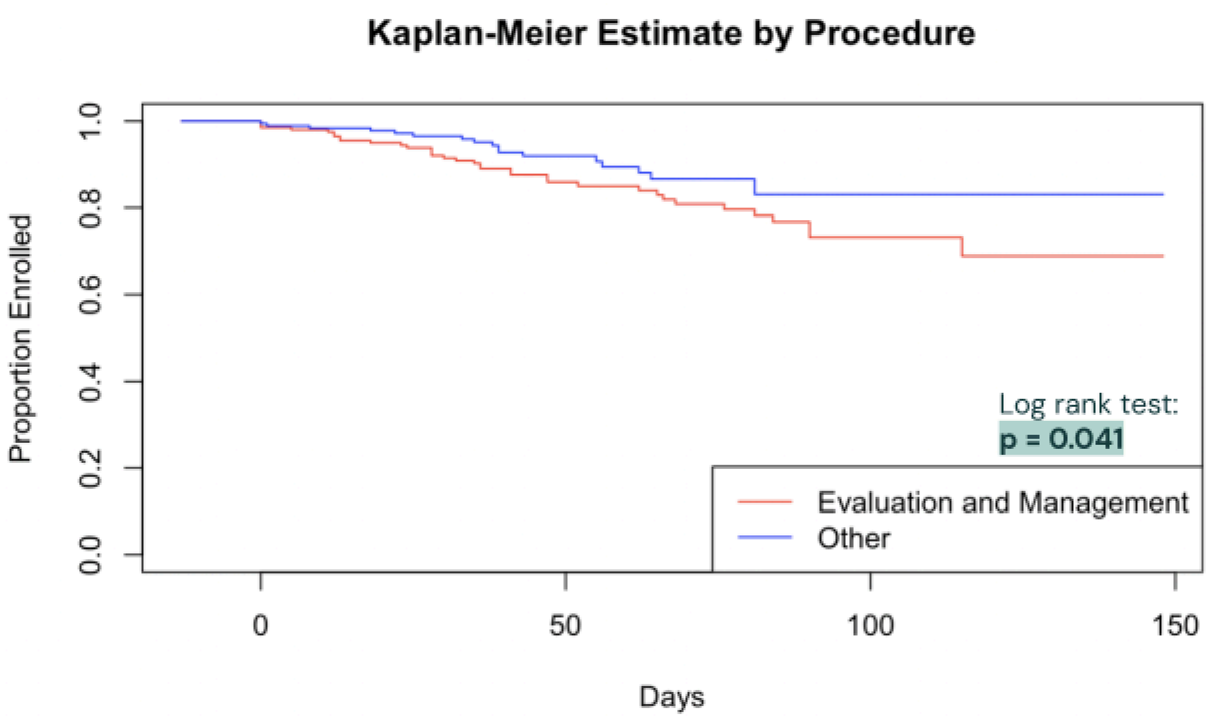


Figure 6



In Figure 5 and 6, the Kaplan-Meier survival curve by procedure group shows the clearest separation in dropout patterns. Because several insurance procedure categories had insufficient sample sizes, they were collapsed into two broader groups: Evaluation and Management versus Other. In the resulting survival curves, the Evaluation and Management group declined more quickly over time, while the Other group remained consistently above it. The log-rank test for this comparison gave a p-value of 0.04, indicating a statistically significant difference in time-to-dropout between the two procedure groups. This suggests that procedure history may be correlated with the chance to dropout.

Model 1: Cox Proportional Hazards Model of Baseline Metrics Variables

```
coxph(formula = Surv(days_enrolled, is_enrolled == 0) ~ mobility +
    mood + sleep + procedure_group + pain_centered + pain_centered_squared,
    data = patients_complete)

  n= 383, number of events= 49

                          coef exp(coef)  se(coef)      z Pr(>|z|)
mobility1             0.123045  1.130936  0.363975  0.338   0.7353
mobility2             0.473223  1.605160  0.567900  0.833   0.4047
mood1                -0.554366  0.574436  0.424795 -1.305   0.1919
mood2                -0.295115  0.744446  0.475696 -0.620   0.5350
sleep1               -0.341896  0.710422  0.343551 -0.995   0.3196
sleep2               -0.337324  0.713677  0.417749 -0.807   0.4194
procedure_groupOther -0.619229  0.538359  0.312598 -1.981   0.0476 *
pain_centered        -0.103219  0.901929  0.072247 -1.429   0.1531
pain_centered_squared -0.000363  0.999637  0.019897 -0.018   0.9854
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

This difference in procedure group also appeared in the Cox proportional hazards model. In Model 1, procedure_groupOther had a coefficient of about -0.62 with a p-value of 0.0476, corresponding to a hazard ratio of about 0.54. This hazard ratio indicates that patients in the Other procedure group had a lower hazard of dropout over time than patients in the Evaluation and Management group. By comparison, the other baseline predictors remained non-significant, including mobility, mood, sleep, pain, and the squared pain term. These results suggest that the procedure grouping was the clearest predictor of dropout timing among the variables examined so far.

Figure 7

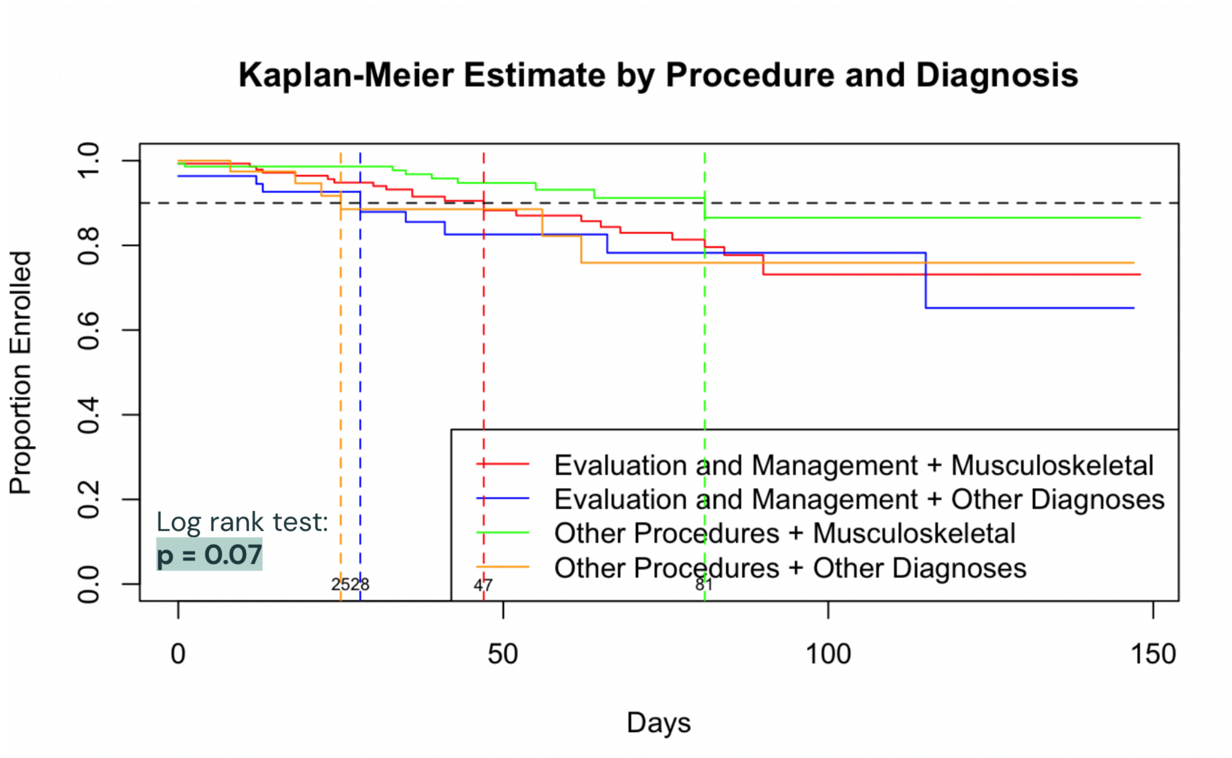


Figure 7 illustrates the proportion of patients remaining enrolled in the study over time, stratified into four distinct groups based on their baseline procedure and diagnosis. The log-rank test p-value of 0.07 indicates a marginally significant difference in dropout rates among these four groups.

Patients who received “Other Procedures” for “Musculoskeletal” diagnoses demonstrated the best overall retention, represented by the green curve that stays highest on the y-axis over time. Patients in this group maintain at least 90% enrollment for approximately 81 days, which is substantially longer than the other groups.

Conversely, patients whose baseline procedure was “Evaluation and Management” tended to drop out at higher rates, regardless of their diagnosis. Visually, these groups correspond to the red and blue lines that step downward more steeply and frequently. These groups fall below 90% enrollment within approximately 28-47 days, indicating much earlier disengagement.

The combination of the baseline procedure and diagnosis chapter provides a nuanced view of patient dropout. The data suggests that the type of procedure is the primary driver of dropout risk, with Evaluation and Management patients dropping out more rapidly. The highest retention is specifically clustered among patients undergoing “Other Procedures” specifically for musculoskeletal issues, suggesting that this specific patient profile may be more invested or required to remain in the program longer.

Model 2: Cox Proportional Hazards Model of Baseline Metrics and Claims Variables

```
coxph(formula = surv_object ~ pain + mobility + mood + sleep +
    procedure_groups + diag_chapter_group, data = patients)

  n= 383, number of events= 49

                            coef exp(coef) se(coef)      z Pr(>|z|)
pain                    -0.10803   0.89760  0.06038 -1.789   0.0736 .
mobility1                0.15043   1.16234  0.36106  0.417   0.6769
mobility2                0.52583   1.69187  0.56274  0.934   0.3501
mood1                   -0.58430   0.55750  0.42302 -1.381   0.1672
mood2                   -0.30646   0.73605  0.47072 -0.651   0.5150
sleep1                  -0.29703   0.74302  0.34721 -0.855   0.3923
sleep2                  -0.31672   0.72853  0.42321 -0.748   0.4542
procedure_groupsOther   -0.57759   0.56125  0.31242 -1.849   0.0645 .
diag_chapter_groupOther  0.53294   1.70393  0.30982  1.720   0.0854 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

In Model 2, the hazard of dropout is regressed on baseline pain score, mobility, mood, sleep, procedure groups, and diagnosis groups. There are three variables showing marginal significance that might indicate weak trends:

1. Pain (p = 0.07): The Hazard Ratio is $e^{-0.11}$ = 0.90. This means that for every 1 unit increase in reported baseline pain, the risk of dropping out decreases by roughly 10%, holding all other variables constant. This suggests a weak trend where patients with more baseline pain might actually stay enrolled in the program slightly longer.

2. Procedure Group (p = 0.06): The Hazard Ratio for the Other procedure group is $e^{-0.58}$ = 0.56. This indicates that patients who had Other procedures have a 44% lower risk of dropping out at any given time compared to the “Evaluation and Management” group, holding all other variables constant.

3. Diagnosis Chapter Group (p = 0.09): The Hazard Ratio for Other diagnoses is $e^{0.53}$ = 1.70. This means patients with Other diagnoses have a 70% higher risk of dropping out compared to the reference group (Diseases of the Musculoskeletal System), holding all other variables constant.

While there are weak trends suggesting that patients with Musculoskeletal diagnoses, Other procedures, and slightly higher baseline pain are more likely to remain in the study, the lack of overall model significance means we cannot confidently state these effects exist in the broader population.

Sentiment Analysis of Patient-MA Messages

Model 3: Logistic Model with Sentiment

```
Call:
glm(formula = dropout ~ pain + mobility + mood + sleep + procedure_groups +
    diag_chapter_group + avg_sentiment, family = "binomial",
    data = patients)

Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)            -1.23589    0.68141  -1.814  0.06972 .
pain                   -0.11475    0.06839  -1.678  0.09337 .
mobility1               0.26826    0.39410   0.681  0.49606
mobility2               0.49918    0.63439   0.787  0.43136
mood1                  -0.73038    0.46191  -1.581  0.11382
mood2                  -0.19210    0.52133  -0.368  0.71251
sleep1                 -0.21997    0.37715  -0.583  0.55974
sleep2                 -0.12256    0.46646  -0.263  0.79275
procedure_groupsOther  -0.62079    0.33654  -1.845  0.06509 .
diag_chapter_groupOther  0.56841    0.34873   1.630  0.10311
avg_sentiment          -1.28918    0.49868  -2.585  0.00973 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

In model 3, we incorporated average patient sentiment into a logistic regression model predicting dropout, while adjusting for the same set of variables.

The results showed that average sentiment was a statistically significant predictor of dropout at alpha=0.05 level. The coefficient for average sentiment was -1.29 with a p-value of 0.0098, indicating that patients with more positive average sentiment were less likely to drop out of the RTM program. In other words, holding the other variables constant, more positive communication was associated with lower odds of disengagement.

Most other predictors were not statistically significant, although there are some weak trends at alpha=0.10 level. Baseline pain showed a negative coefficient (coefficient = -0.115, p = 0.093), suggesting that patients with higher baseline pain are somewhat less likely to drop out. The Other procedure group also showed a negative coefficient (coefficient = -0.621, p = 0.065), consistent with earlier findings that these patients tend to remain enrolled longer than those in the Evaluation and Management group. Overall, the logistic regression results suggest that sentiment provides meaningful additional information for predicting dropouts.

Incorporating sentiment scores into the logistic regression model also improved overall model performance and provided additional predictive value beyond baseline metrics and claims variables.

Table 1: ANOVA Model Comparison

```
Analysis of Deviance Table

Model 1: dropout ~ pain + mobility + mood + sleep + procedure_groups +
    diag_chapter_group + avg_sentiment
Model 2: dropout ~ pain + mobility + mood + sleep + procedure_groups +
    diag_chapter_group
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       372     272.02
2       373     279.36 -1  -7.3429 0.006733 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

First, as shown in Table 1, model comparison using ANOVA indicated that the inclusion of sentiment significantly improved model fit (p = 0.00673), This indicates that sentiment contributes additional explanatory power beyond the baseline variables.

Table 2: AIC Comparison

	df <dbl>	AIC <dbl>
without_sentiment	10	299.3609
with_sentiment	11	294.0180

Second, Table 2 comparison of Akaike Information Criterion (AIC) values further supported this improvement. The model including sentiment achieved a lower AIC (294.02) compared to the model without sentiment (299.36). Since lower AIC values indicate better model fit after penalizing for model complexity, this further supports the inclusion of sentiment as improving overall model performance.

Table 3: Hosmer–Lemeshow Goodness-of-Fit (p-value = 0.72)

Group	N	Observed	Expected	Avg_p_hat	Chi_Sq
1	39	0	1.17	0.030	1.167
2	39	1	2.01	0.052	0.508
3	39	2	2.62	0.067	0.146
4	38	5	3.22	0.085	0.990
5	38	3	3.91	0.103	0.212
6	38	6	4.67	0.123	0.382
7	38	7	5.45	0.143	0.442
8	38	8	6.71	0.177	0.248
9	38	9	7.92	0.208	0.148
10	38	8	11.34	0.298	0.982

Finally, the Hosmer–Lemeshow goodness-of-fit test showed no evidence of poor calibration ($p = 0.72$), suggesting that the model's predicted probabilities align well with observed dropout outcomes. This is further supported by the observed versus expected dropout counts across risk deciles shown in Table 3, which demonstrate reasonable agreement between predicted and actual outcomes.

Overall, all three evaluation metrics consistently indicate that sentiment is a valuable predictor and enhances the model's ability to accurately estimate patient dropout risk.

Discussion

Overall, the exploratory analysis suggests that patient engagement in the RTM program may be related to how pain levels evolve over time. The individual patterns suggest that patients who perceive ongoing benefit from monitoring their condition may be more likely to remain engaged, while patients whose symptoms quickly resolve may be less motivated to continue participation. Although these observations are based on a small number of cases, they highlight potential behavioral and health-related signals that could be useful for predicting patient drop-off. In addition to these trends, our updated analysis introduces sentiment derived from patient-provider communication as a meaningful behavioral signal, where more positive sentiment is associated with a lower likelihood of dropout, suggesting that perceived experience and communication quality may also play an important role in sustained engagement.

The baseline comparison graph shows the variance in the patient's pain level on the first day of the program (Figure 3). By bare eyes, it suggests a very weak relationship that patients with higher baseline pain level are more likely to stay in the program. This aligns with our broader findings that baseline clinical variables alone may not be sufficient predictors of engagement, especially when compared to more dynamic or experiential measures such as sentiment.

The current EDA provides a useful starting point for understanding patient disengagement in Flagler Health's RTM program, but it also reveals several important limitations. So far, the analysis has focused solely on the ordinal, categorical health metrics collected during patient check-ins. While these variables are valuable because they are relatively straightforward and directly connected to patients' health conditions, they do not capture the full patient experience, especially the more qualitative and behavioral aspects of engagement. In particular, the messages and insurance data remain largely unexplored, even though disengagement may also be influenced by communication patterns, patient responsiveness, and treatment history. Our newer results begin to address this gap by incorporating sentiment from chat data, which was found to be a statistically significant predictor of dropout (coefficient = −1.29, p-value = 0.01), reinforcing the importance of integrating qualitative data into the analysis.

Building on this limitation, our updated analysis suggests that insurance-based treatment context may provide more useful information about disengagement than the baseline health metrics alone. The survival analysis showed that baseline pain, mobility, mood, and sleep were not strong predictors of dropout, while the procedure group showed the clearest difference in retention patterns. After collapsing the small sample sized procedure categories into Evaluation and Management versus Other, we found that patients in the Other groups tended to remain enrolled longer, and this result was consistent across the Kaplan-Meier curves and Cox model. An educated guess is that the Evaluation and Management group is often tied to one time medical consultation, and patients might just want to take the actual therapy process somewhere else. In contrast, surgery related care may reflect a more long-term relationship with care, as patients recovering from surgery might also take follow-up care in the same clinic, so the dropped rate is lower compared to patients in the Evaluation and Management group. This finding is further supported by our modeling results, where procedure grouping consistently showed clear differences in dropout patterns and appears to capture structural aspects of patient engagement, while sentiment provides an additional layer of insight into patient-level variation within these groups.

Our analysis incorporating diagnosis groups provided an even more nuanced view of patient dropoff. Specifically, patients undergoing “Other Procedures” for musculoskeletal issues demonstrated the highest retention, whereas Evaluation and Management patients dropped out fastest regardless of diagnosis, suggesting the program is currently better suited for active physical rehabilitation rather than standard consults. Additionally, while mobility, mood, and sleep showed no predictive value, the survival model indicated a weak but clinically intuitive trend where higher initial pain slightly decreased the risk of dropout. However, due to the small number of dropouts and the lack of strict statistical significance in the overall model, these findings remain hypothesis-generating. At the same time, the inclusion of sentiment suggests that even within similar diagnosis or procedure groups, patients with more negative communication patterns may still be at higher risk of disengagement.

Since the baseline variables did not appear to have strong predictive value, our next steps could be to explore Repeated Measures Logistic Regression. This would allow us to move beyond only baseline predictors and instead model how changes in patient metrics over time relate to dropout. Potential features could include within-patient trends in pain, changes from baseline, variability in metrics across visits, frequency of medical insurance logs, and measures taken right before dropout rather than only at baseline, right after enrollment. For example, if a patient’s pain decreases quickly, they may feel that they no longer need the program. It may also be useful to further explore the text data by extracting features such as summarized patient and medical assistant sentiment, sentiment trajectories over time, response rates, and the sentiment of the last message before dropout. In addition, interaction effects, such as between sentiment and procedure group, could help capture more complex relationships in engagement. Finally, the diagnosis codes could be refined into smaller and more meaningful subcategories, since most patients currently fall under the broad category of “Diseases of the Musculoskeletal System and Connective Tissue” because the RTM program primarily serves musculoskeletal health patients.

Appendix A

Insurance Groupings

Figure A1

Categorization of ICD-10 Diagnosis Codes

Code.Range <chr>	Description <chr>
A00–B99	Certain Infectious and Parasitic Diseases
C00–D49	Neoplasms
D50–D89	Diseases of the Blood and Blood–Forming Organs and Certain Disorders Involving the Immune Mechanism
E00–E89	Endocrine, Nutritional and Metabolic Diseases
F01–F99	Mental, Behavioral and Neurodevelopmental Disorders
G00–G99	Diseases of the Nervous System
H00–H59	Diseases of the Eye and Adnexa
H60–H95	Diseases of the Ear and Mastoid Process
I00–I99	Diseases of the Circulatory System
J00–J99	Diseases of the Respiratory System
K00–K95	Diseases of the Digestive System
L00–L99	Diseases of the Skin and Subcutaneous Tissue
M00–M99	Diseases of the Musculoskeletal System and Connective Tissue
N00–N99	Diseases of the Genitourinary System
O00–O9A	Pregnancy, Childbirth and the Puerperium
P00–P96	Certain Conditions Originating in the Perinatal Period
Q00–Q99	Congenital Malformations, Deformations and Chromosomal Abnormalities
R00–R99	Symptoms, Signs, and Abnormal Clinical and Laboratory Findings, Not Elsewhere Classified
S00–T88	Injury, Poisoning, and Certain Other Consequences of External Causes
U00–U85	Codes for Special Purposes
V00–Y99	External Causes of Morbidity
Z00–Z99	Factors Influencing Health Status and Contact with Health Services

Note: Categories referenced from this website – https://www.aapc.com/resources/what-is-icd-10

Figure A2

Categorization of CPT I Procedure Codes

Key.Category <chr>	CPT.Code.Range.s. <chr>
Evaluation & Management (E/M)	99202–99499
Anesthesia	00100–01999, 99100–99140
Surgery	10004–69990
Radiology	70010–79999
Pathology & Laboratory	80047–89398
Medicine	90281–99199, 99500–99607

Note: Categories referenced from this website – https://www.aapc.com/resources/what-is-cpt

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